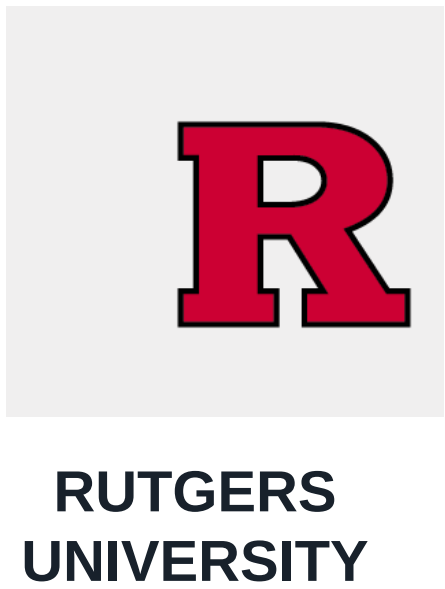


TEMPORAL VISUAL REUSE IN OPENVLA-OFT

A sequential negative study of caching across robot-control queries

Ved Dwivedi¹ | Cheng Yang² | Bo Yuan²

¹ John P. Stevens High School ² Department of Electrical and Computer Engineering, Rutgers University-New Brunswick



RESEARCH QUESTION

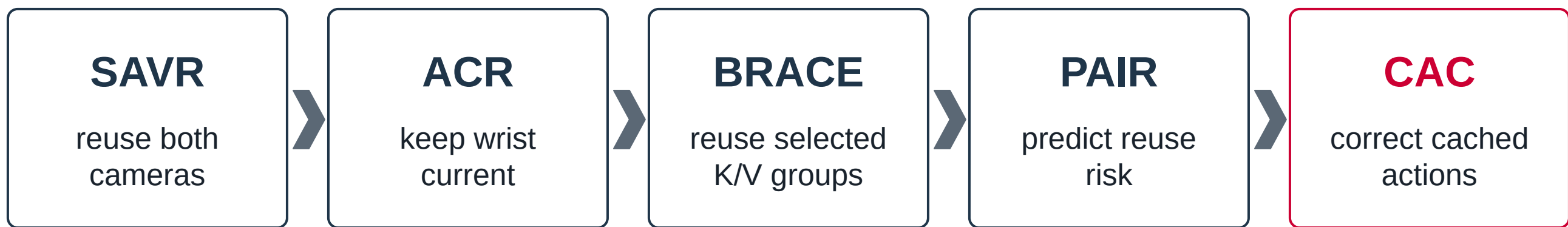
Can a frozen robot policy reuse visual computation across queries without losing task success?

SHORT ANSWER

Not under the tested designs: savings and reliability did not coexist.

FIVE INCREASINGLY TARGETED TESTS

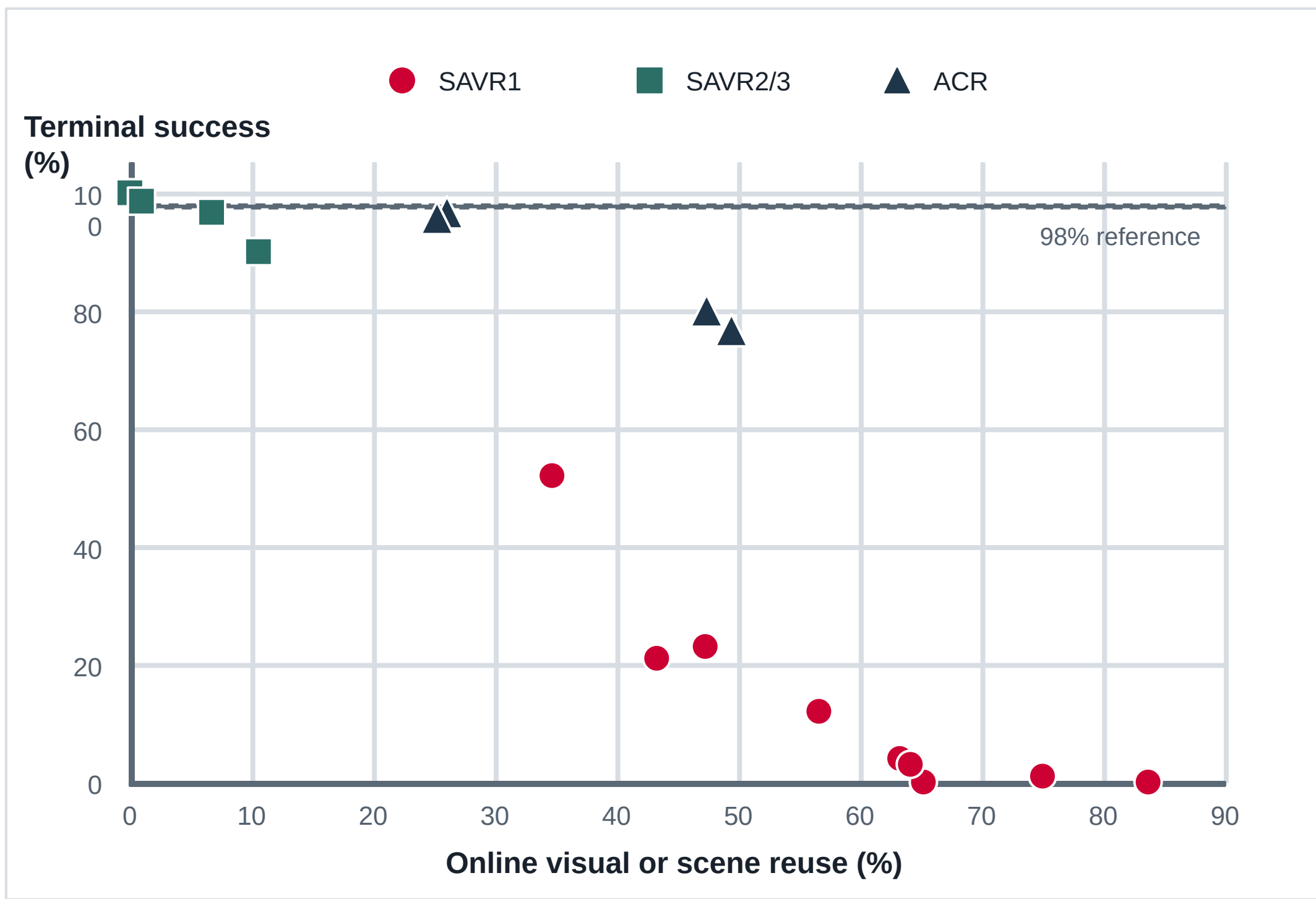
Each redesign addressed the failure observed in the preceding stage.



Test bed: OpenVLA-OFT on LIBERO | terminal task success + complete-query timing | frozen gates and independent semantic checks

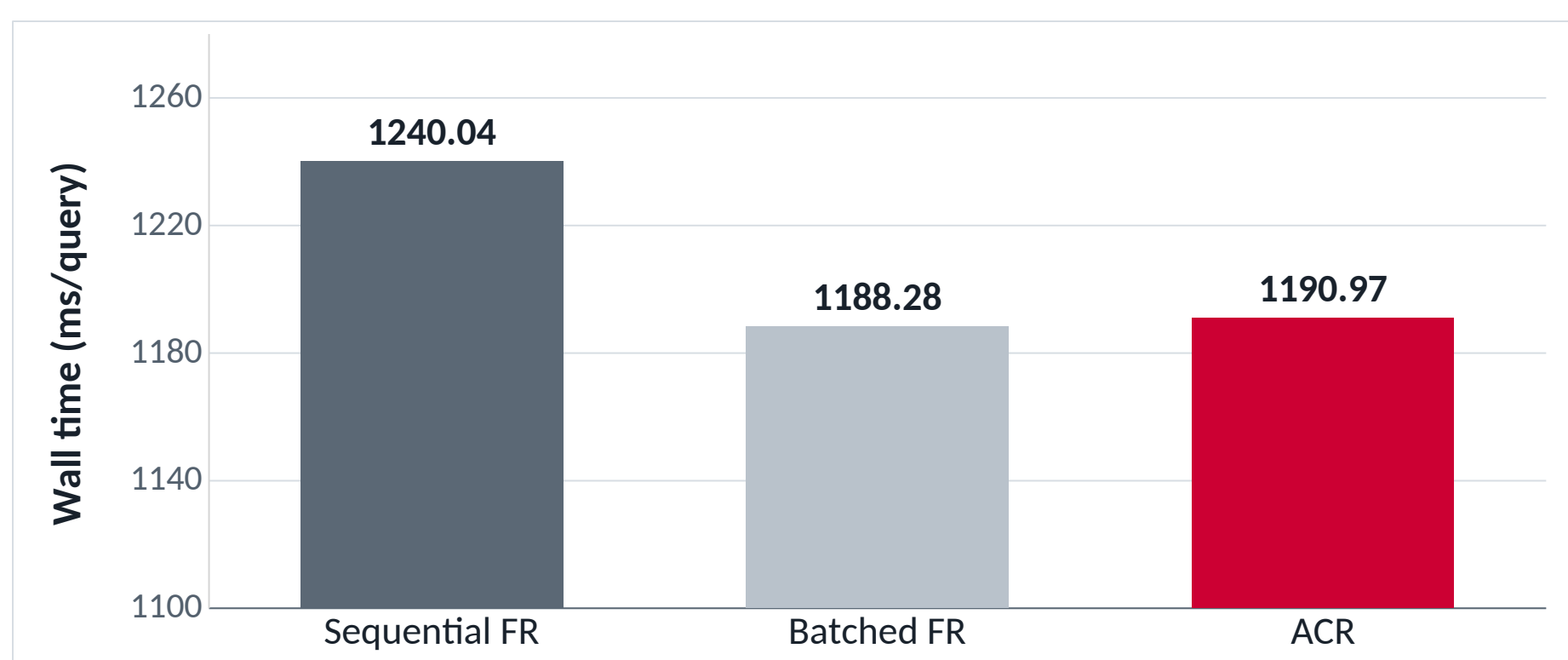
RESULTS: COMPUTE SAVINGS APPEARED, BUT NO METHOD PASSED BOTH GATES

1 More reuse generally reduced success



High-success safeguards worked only by suppressing reuse. The strongest SAVR redesign reached 69/70 success but reused just 0.95% of visual queries.

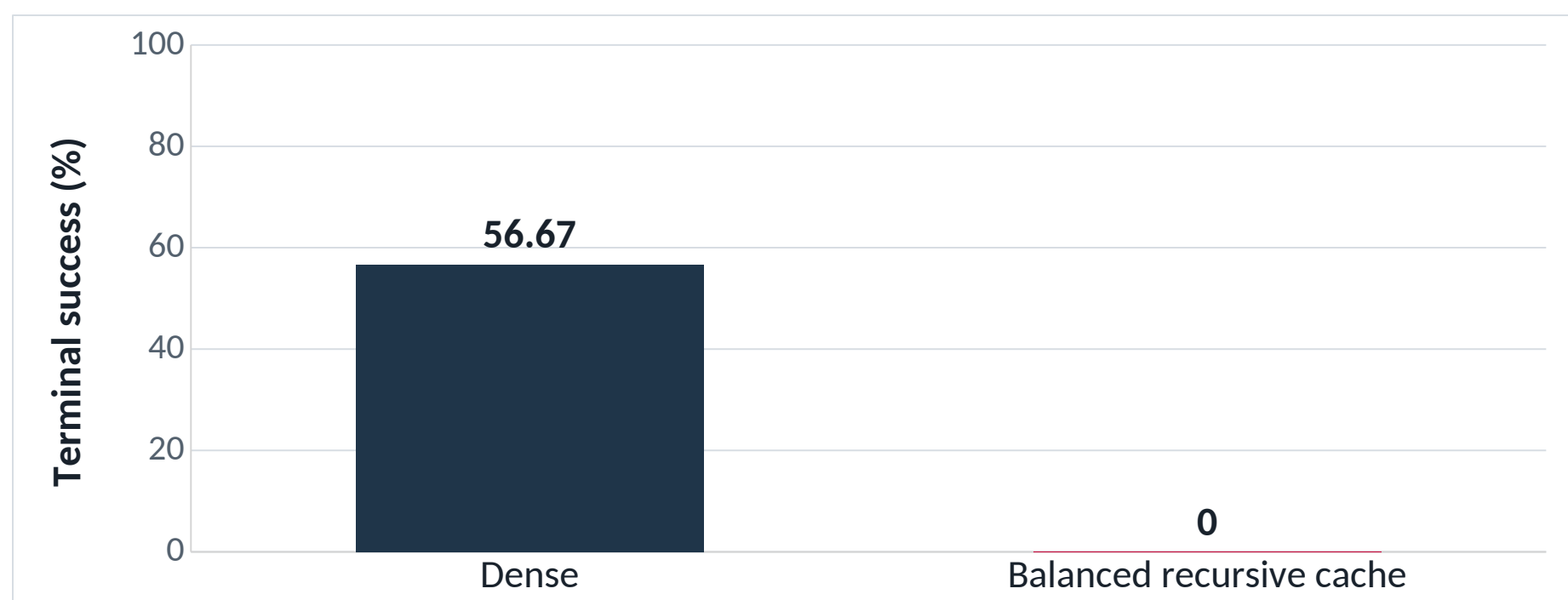
2 Camera asymmetry preserved success - not speed



67/70 = 67/70

ACR success = batched full refresh

3 The final cache substrate failed closed-loop control



Dense: 68/120 successes | Cache: 0/120 | 22.60% gross timing headroom in systems qualification

CORE FINDING | Useful compute savings appeared, but no tested frozen-model method achieved both reliability and incremental speed.

WHY CLOSED-LOOP REUSE IS HARD

Recursive reuse combines internal states from different physical times. A changed action changes the next observation, so offline replay misses the feedback loop.



feedback compounds across control queries



WHAT THE STUDY CONTRIBUTES

- Optimize the fresh baseline first**
Batching erased ACR's apparent speedup.
- Check model semantics independently**
Shared helper code hid a one-token action-readout error.
- Test closed loop early**
Offline similarity and risk signals did not guarantee control.
- Next direction**
Preserve current task-critical information or train the model under reuse.

SCOPE | One OpenVLA-OFT checkpoint, one simulator benchmark family, one GPU class. Cross-stage comparisons are descriptive, not randomized head-to-head estimates.

Key references: OpenVLA (CoRL 2024); OpenVLA-OFT (RSS 2025); VLA-Cache (NeurIPS 2025); LIBERO (NeurIPS 2023).



CODE + ARTIFACTS
github.com/vwdr/SAVR